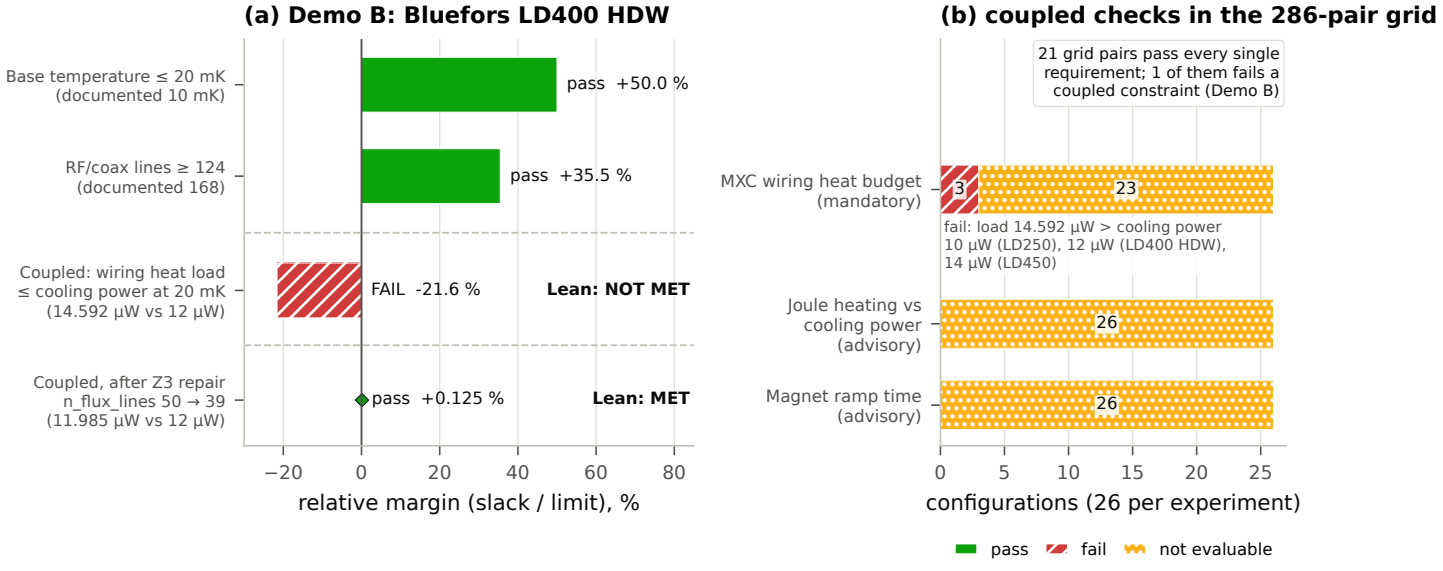

(a) Demo B: Bluefors LD400 HDW
Base temperature ≤ 20 mK
(documented 10 mK)
pass +50.0 %
RF/coax lines ≥ 124
(documented 168)
pass +35.5 %
Coupled: wiring heat load
≤ cooling power at 20 mK
(14.592 µW vs 12 µW)
FAIL -21.6 %
Lean: NOT MET
Coupled, after Z3 repair
n_flux_lines 50 → 39
(11.985 µW vs 12 µW)
pass +0.125 %
Lean: MET
−20
0
20
40
60
80
relative margin (slack / limit), %
(b) coupled checks in the 286-pair grid
21 grid pairs pass every single
requirement; 1 of them fails a
coupled constraint (Demo B)
MXC wiring heat budget
(mandatory)
3
23
fail: load 14.592 µW > cooling power
10 µW (LD250), 12 µW (LD400 HDW),
14 µW (LD450)
Joule heating vs
cooling power
(advisory)
26
Magnet ramp time
(advisory)
26
0
5
10
15
20
25
configurations (26 per experiment)
pass
fail
not evaluable